



Section 2.7 – Worked Examples

*Examples 2.22 – 2.27 : Modulus, Argument, Polar Form,
Product & Quotient using the Properties of Polar Form*





Properties of Polar Form

Three key rules we will use in the examples ahead



~~Property 1 — Inverse of z~~

Property 1 If $z = r(\cos \theta + i \sin \theta)$, then $z^{-1} = (1/r)(\cos \theta - i \sin \theta)$

Proof (short):

$$z^{-1} = 1/z = 1 / [r(\cos \theta + i \sin \theta)]$$

Multiply top & bottom by $(\cos \theta - i \sin \theta)$:

$$z^{-1} = (\cos \theta - i \sin \theta) / [r(\cos^2 \theta + \sin^2 \theta)]$$

Since $\cos^2 \theta + \sin^2 \theta = 1$:

$$z^{-1} = (1/r)(\cos \theta - i \sin \theta)$$

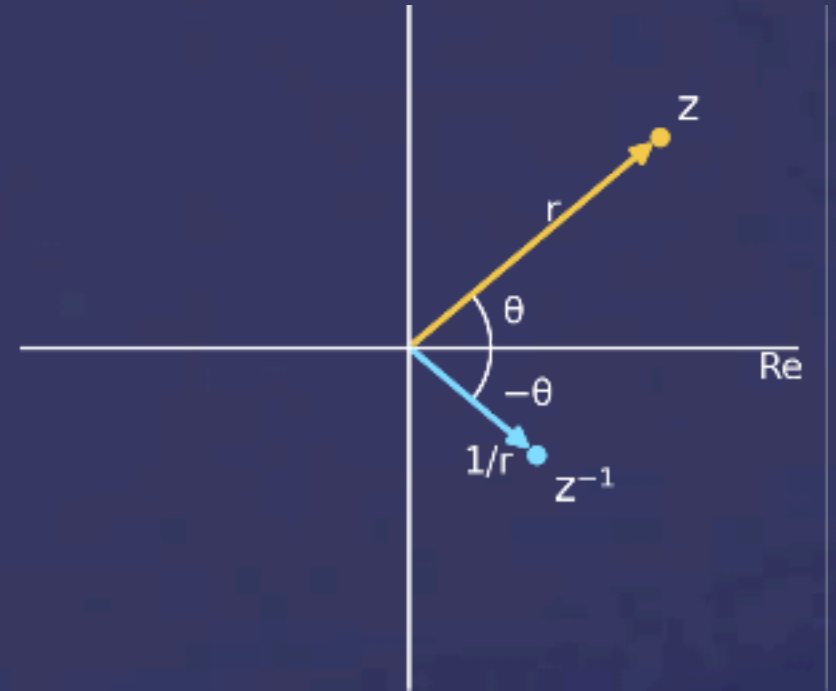


Fig 2.41 : z and z^{-1} are reflected in the real axis, radius scales by $1/r$

Property 2 — Product of z_1 and z_2

Property 2 If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$, then $z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2)]$

Idea:

Multiply the two polar forms and expand.

Using $\cos(A+B)$ and $\sin(A+B)$ formulas, the product simplifies to:

$$z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$$

Note: $\arg(z_1 z_2) = \theta_1 + \theta_2 = \arg(z_1) + \arg(z_2)$

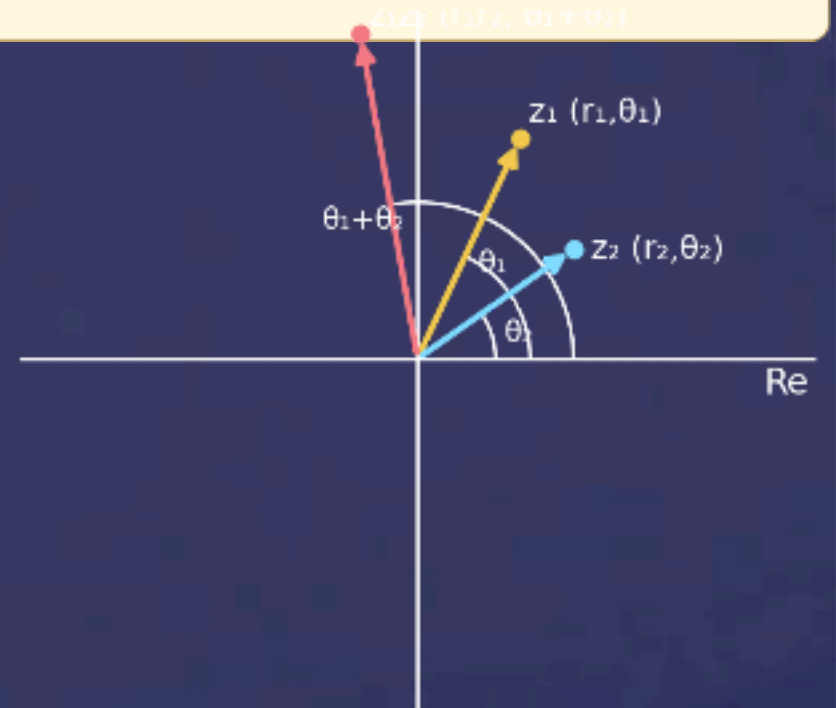


Fig 2.42 : multiply moduli, add arguments

Property 3 — Quotient of z_1 and z_2

Property 3 $z_1/z_2 = (r_1/r_2) [\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$

Idea:

Multiply top & bottom of z_1/z_2 by $(\cos \theta_2 - i \sin \theta_2)$, then simplify.

$$z_1/z_2 = (r_1/r_2)[\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$$

Note: $\arg(z_1/z_2) = \theta_1 - \theta_2 = \arg(z_1) - \arg(z_2)$

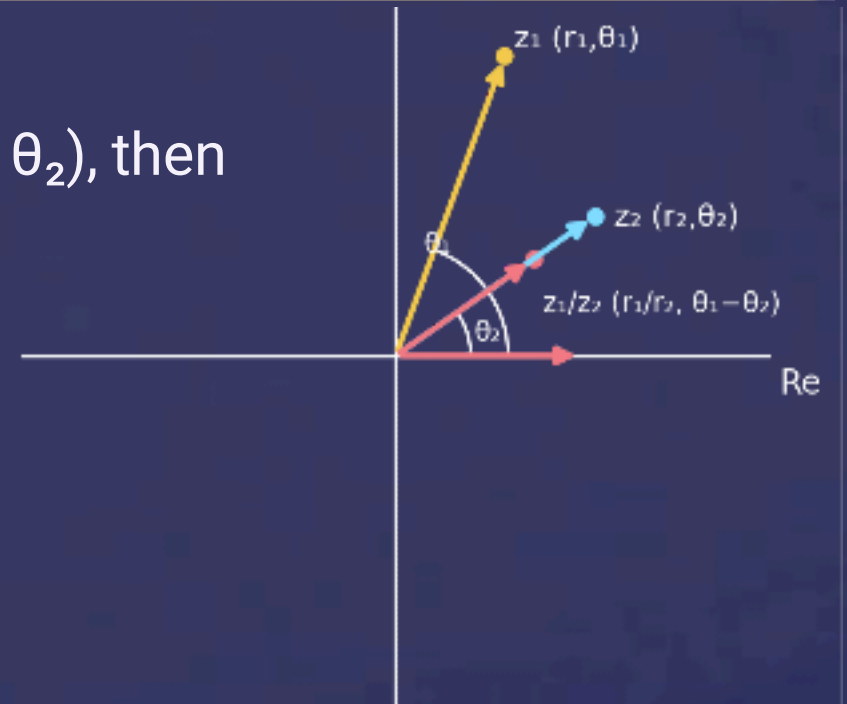


Fig 2.43 : divide moduli, subtract arguments



Example 2.22

*Find the modulus and principal argument of
 $\sqrt{3}+j$, $-\sqrt{3}+j$, $-\sqrt{3}-j$, $\sqrt{3}-j$*



Example 2.22 — parts (i) and (ii)

(i) $\sqrt{3} + i$

Modulus = $\sqrt{x^2+y^2} = \sqrt{((\sqrt{3})^2 + 1^2)} = \sqrt{4} = 2$

$\alpha = \tan^{-1}|y/x| = \tan^{-1}(1/\sqrt{3}) = \pi/6$

Lies in Quadrant I $\Rightarrow \theta = \alpha = \pi/6$

Modulus = 2, Principal argument = $\pi/6$

(ii) $-\sqrt{3} + i$

Modulus = 2, $\alpha = \pi/6$ (same reference angle)

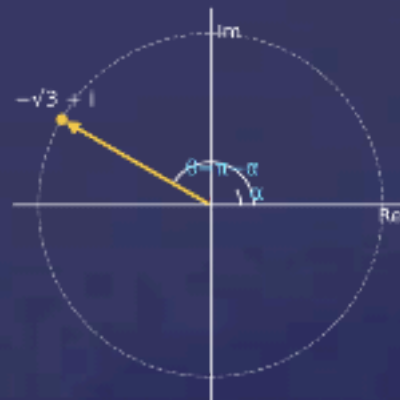
Lies in Quadrant II $\Rightarrow \theta = \pi - \alpha = \pi - \pi/6 = 5\pi/6$

Modulus = 2, Principal argument = $5\pi/6$

Fig 2.36 : $\sqrt{3} + i$ (Quadrant I)



Fig 2.37 : $-\sqrt{3} + i$ (Quadrant II)



Example 2.22 – parts (iii), (iv) and conclusion

(iii) $-\sqrt{3} - i$

$r = 2$, $\alpha = \pi/6$. Lies in Quadrant III

$$\Rightarrow \theta = \alpha - \pi = \pi/6 - \pi = -5\pi/6$$

Modulus = 2, Principal argument = $-5\pi/6$

(iv) $\sqrt{3} - i$

$r = 2$, $\alpha = \pi/6$. Lies in Quadrant IV

$$\Rightarrow \theta = -\alpha = -\pi/6$$

Modulus = 2, Principal argument = $-\pi/6$

Fig 2.38 : $-\sqrt{3} - i$ (Quadrant III)

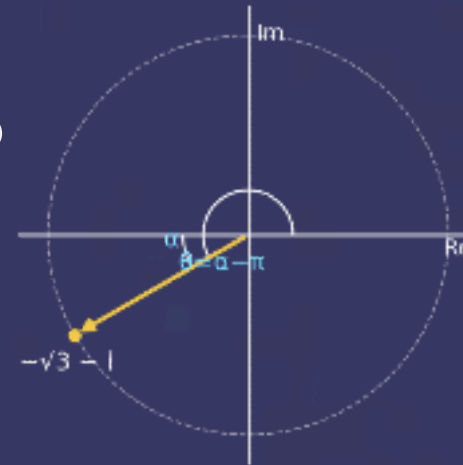
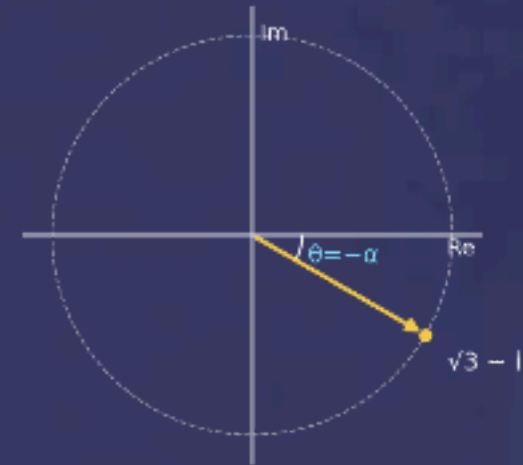


Fig 2.39 : $\sqrt{3} - i$ (Quadrant IV)



Conclusion: in all four cases the modulus is equal (2), but the argument depends on which quadrant the complex number lies in.





Example 2.23

Represent $-1 - i$ and $1 + i\sqrt{3}$ in polar form



Example 2.23 (i) — Polar form of $-1 - i$

Let $-1 - i = r(\cos \theta + i \sin \theta)$

$$r = \sqrt{x^2 + y^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\alpha = \tan^{-1}|y/x| = \tan^{-1}(1) = \pi/4$$

$-1 - i$ lies in Quadrant III $\Rightarrow \theta = \alpha - \pi = \pi/4 - \pi = -3\pi/4$

$$\begin{aligned}\text{So: } -1 - i &= \sqrt{2} [\cos(-3\pi/4) + i \sin(-3\pi/4)] \\ &= \sqrt{2} [\cos(3\pi/4) - i \sin(3\pi/4)]\end{aligned}$$

General form (all values of k):

$$-1 - i = \sqrt{2} [\cos(3\pi/4 + 2k\pi) - i \sin(3\pi/4 + 2k\pi)], \quad k \in \mathbb{Z}$$

Note: different k values give different alternative polar forms.



Example 2.23 (ii) — Polar form of $1 + i\sqrt{3}$

$$r = |z| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

$$\theta = \tan^{-1}(\sqrt{3}/1) = \tan^{-1}(\sqrt{3}) = \pi/3$$

$$\text{Hence } \arg(z) = \pi/3$$

Therefore, the polar form is:

$$1 + i\sqrt{3} = 2 (\cos \pi/3 + i \sin \pi/3)$$

General form:

$$1 + i\sqrt{3} = 2 [\cos(\pi/3 + 2k\pi) + i \sin(\pi/3 + 2k\pi)], \quad k \in \mathbb{Z}$$





Example 2.24

Find the principal argument $\text{Arg } z$, when $z = -2 / (1 + i\sqrt{3})$



Example 2.24 — Principal argument of a quotient

Uses: $\arg(z_1/z_2) = \arg z_1 - \arg z_2$

$$\begin{aligned}\arg z &= \arg[-2 / (1+i\sqrt{3})] \\ &= \arg(-2) - \arg(1+i\sqrt{3}) \\ &= [\pi - \tan^{-1}(0/2)] - \tan^{-1}(\sqrt{3}/1) \\ &= \pi - \pi/3 = 2\pi/3\end{aligned}$$

So one value of $\arg z$ is $2\pi/3$.
Since $2\pi/3$ lies between $-\pi$ and π ,
the principal argument $\text{Arg } z = 2\pi/3$.

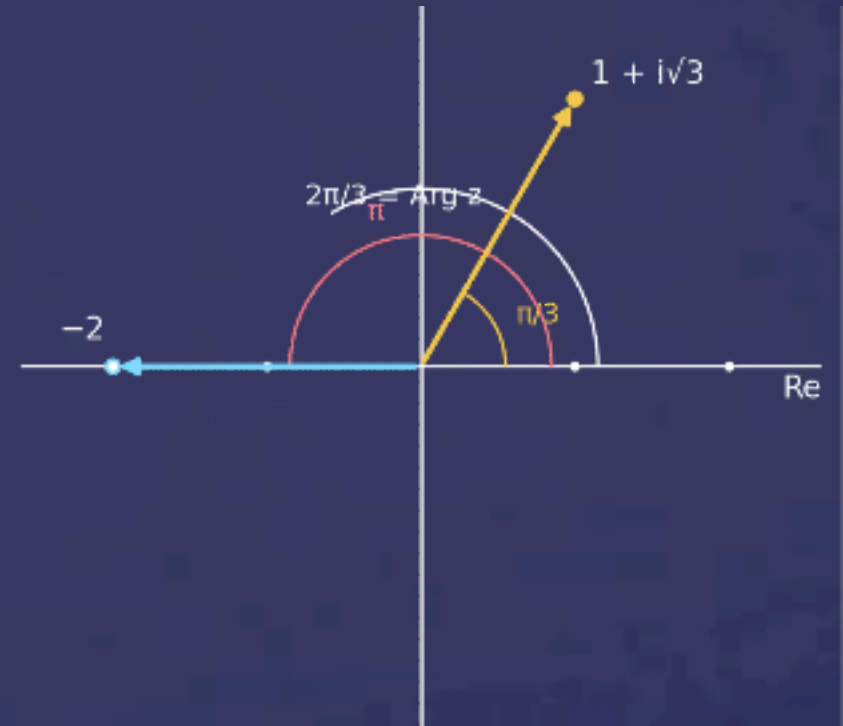


Fig 2.40



Example 2.25

Find the product
 $(3/2)(\cos \pi/3 + i \sin \pi/3) \cdot 6(\cos 5\pi/6 + i \sin 5\pi/6)$
in rectangular form



Example 2.25 — Product in rectangular form

Uses Property 2: $r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$

Multiply the moduli and add the arguments:

$$= (3/2)(6) [\cos(\pi/3 + 5\pi/6) + i \sin(\pi/3 + 5\pi/6)]$$

$$= 9 [\cos(7\pi/6) + i \sin(7\pi/6)]$$

$$= 9 [\cos(\pi + \pi/6) + i \sin(\pi + \pi/6)]$$

$$= 9 [-\cos(\pi/6) - i \sin(\pi/6)]$$

$$= 9 (-\sqrt{3}/2 - i/2)$$

$$= -9\sqrt{3}/2 - 9i/2$$

This is the rectangular form.





Example 2.26

Find the quotient

$$2(\cos 9\pi/4 + i \sin 9\pi/4) \div 4(\cos(-3\pi/2) + i \sin(-3\pi/2))$$

in rectangular form





Example 2.26 – Quotient, step 1: subtract arguments

Uses Property 3: $(r_1/r_2)[\cos(\theta_1-\theta_2)+i\sin(\theta_1-\theta_2)]$

Divide moduli, subtract arguments:

$$= (2/4) [\cos(9\pi/4 - (-3\pi/2)) + i \sin(9\pi/4 - (-3\pi/2))]$$

$$= (1/2) [\cos(9\pi/4 + 3\pi/2) + i \sin(9\pi/4 + 3\pi/2)]$$

$$= (1/2) [\cos(15\pi/4) + i \sin(15\pi/4)]$$

$15\pi/4 = 4\pi - \pi/4$, and $\cos/\sin$ repeat every 2π (here every 4π works too):

$$= (1/2) [\cos(4\pi - \pi/4) + i \sin(4\pi - \pi/4)]$$

$$= (1/2) [\cos(\pi/4) - i \sin(\pi/4)]$$

Example 2.26 — Quotient, step 2: simplify to rectangular form

$$\begin{aligned} &= (1/2)(1/\sqrt{2} - i \cdot 1/\sqrt{2}) \\ &= 1/(2\sqrt{2}) - i \cdot 1/(2\sqrt{2}) \\ &= \sqrt{2}/4 - i \cdot \sqrt{2}/4 \end{aligned}$$

This is the rectangular form of the given quotient.





Example 2.27

*If $z = x + iy$ and $\arg[(z-1)/(z+1)] = \pi/2$,
show that $x^2 + y^2 = 1$*



Example 2.27 — Proof

$$(z-1)/(z+1) = (x+iy-1)/(x+iy+1) = [(x-1)+iy] / [(x+1)+iy]$$

Multiply top & bottom by the conjugate of the denominator:

$$= [(x-1)+iy][(x+1)-iy] / [(x+1)+iy][(x+1)-iy]$$

$$(z-1)/(z+1) = [(x^2+y^2-1) + i(2y)] / [(x+1)^2 + y^2]$$

$$\text{Given: } \arg[(z-1)/(z+1)] = \pi/2$$

$$\Rightarrow \tan^{-1}[2y / (x^2+y^2-1)] = \pi/2$$

$$\Rightarrow 2y/(x^2+y^2-1) = \tan(\pi/2) \Rightarrow x^2+y^2-1 = 0$$

$$\Rightarrow x^2 + y^2 = 1 \quad \checkmark \text{ Proved}$$

